

Head-to-head: priority-fusion vs steady-aim-cascade, both re-tuned on the full corpus

Results

Both lineages were re-tuned from scratch — 2,000 trials, 1,000 each — on all 531,431 rows of 1-train, under one loader, one 58-label catalogue, one fit/calibration curve and one fixed evaluator, then scored on all eight sealed 2-eval corpora (126,129 rows).

Model scoreboard

Measure	fusion-1k (arm A)	steady-cascade (arm B)	fusion-12k (arm C)
Read window (chars)	1,000	12,000	12,000
macro F2 (headline ranker)	0.4800	0.6497	0.4809
macro F1	0.4066	0.6235	0.4037
macro F0.5	0.3712	0.6147	0.3663
macro F3	0.5308	0.6682	0.5344
macro precision	0.3573	0.6154	0.3512
macro recall	0.7946	0.6482	0.8071
micro F1	0.3652	0.7313	0.3647
micro F2	0.5443	0.8025	0.5459
micro F0.5	0.2797	0.6730	0.2788
micro precision	0.2429	0.6397	0.2418
micro recall	0.9031	0.8598	0.9152
priority macro F0.5 (contra ranker)	0.2060	0.7467	0.2054
priority macro precision	0.1829	0.7478	0.1824
priority macro recall	0.9991	0.7622	0.9991
worst priority-tag recall	0.9966	0.6524	0.9974
document precision	0.6163	0.8893	0.6163
document recall	1.0000	0.7532	1.0000
document specificity	0.0006	0.8832	0.0007
prediction rate	0.9999	0.7836	0.9999
dead tag×corpus pairs (F2 = 0)	8	16	6

Measure	fusion-1k (arm A)	steady-cascade (arm B)	fusion-12k (arm C)
median tag F2	0.4004	0.6667	0.3971
one-core p95 (ms)	1.157	3.916	4.232
one-core docs/s	1361	314	270

The cascade wins every summary metric that prices precision at all — including macro F2, the metric chosen because it favours recall-first models. It leads on all five corpora whose gold can measure F2, with non-overlapping 95% intervals on every one:

Corpus	n	fusion-1k	steady-cascade	fusion-12k
betterdataai_ner_silver	10,360	0.2485 [0.1719, 0.2728]	0.3122 [0.2603, 0.3275]	0.2485
ai4privacy_pii_masking	10,626	0.5238 [0.5158, 0.5313]	0.7476 [0.7386, 0.7560]	0.5242
pii_holdout	20,000	0.5512 [0.5464, 0.5563]	0.7411 [0.7272, 0.7530]	0.5699
pii2_eval	30,000	0.4808 [0.4749, 0.4859]	0.7822 [0.7783, 0.7858]	0.4659
openpii_pii_eval	38,937	0.5957 [0.5881, 0.6067]	0.6656 [0.6644, 0.6669]	0.5962
datax · nemotron · govdocs2	15,206	<i>NOT MEASURABLE</i>	<i>NOT MEASURABLE</i>	<i>NOT MEASURABLE</i>
equal-corpus mean		0.4800	0.6497	0.4809

Three of the eight corpora carry positive-only or partial tag gold, so no precision-bearing metric is computable on them. They report *NOT MEASURABLE*, never 0.0, and are excluded from the mean.

The decisions, both declared before the numbers

Headline policy (per-priority-tag recall ≥ 0.90 conclusive; one-core p95 ≤ 5 ms; ranked on macro F2):

Arm	Verdict	Hard constraints failed
arm-A	FEASIBLE	0 of 2
arm-B	blocked	1 of 2 — priority tag recall: 25 of 55 measurable pairs passed
arm-C	WINNER	0 of 2

Contra-view policy (document precision ≥ 0.90 , specificity ≥ 0.85 , recall ≥ 0.85 , per-priority-tag recall ≥ 0.75 , all conclusive; ranked on priority macro F0.5):

Arm	Verdict	Hard constraints failed
arm-A	blocked	2 of 5 — doc precision 0/3 pairs, doc specificity 0/3 pairs
arm-B	blocked	3 of 5 — doc precision 1/3, doc recall 1/3, priority recall 43/55
arm-C	blocked	2 of 5 — doc precision 0/3 pairs, doc specificity 0/3 pairs

No arm ships under the contra-view policy. Selected under the headline policy: **arm-C**, by 0.0009 macro F2 over arm-A — which is not a real ordering; see below.

The finding that matters most

The fusion arms clear the 0.90 recall gate by saturation, not by capability. Their document specificity is **0.0006** — across 10,003 documents that judges confirmed contain no PII, the fusion stays silent on about **six**. Its prediction rate is 0.9999. A model that says "this contains PII" about essentially every document has a per-tag recall near 1.0 by construction, and that is exactly the degenerate optimum the profile's advisory metrics exist to expose.

Read the two columns together:

	fusion (A/C)	cascade (B)
priority macro recall	0.9991	0.7622
priority macro precision	0.1829	0.7478
document specificity	0.0006	0.8832

The gate is satisfied and the model is not useful. The cascade fails the same gate honestly: 30 of 55 priority tag×corpus pairs fall below 0.90, and at the contra-view's 0.75 floor only 12 of 55 still fail. Its worst pairs are real gaps, not thin scopes — `sensitive_pii_address@govdocs2` at 0.5252 (n=891), `sensitive_pii_full_name@govdocs2` at 0.5493 (n=3,215).

TL;DR

- **Both models were re-tuned from scratch on the full 531,431-row corpus** — 2,000 trials total — under one loader, one catalogue, one fit/calibration curve (451,548 / 79,883, identical rows for both) and one fixed evaluator.
- **The steady-aim cascade is the better model, decisively.** macro F2 **0.6497 vs 0.4809**, micro F1 **0.7313 vs 0.3647**, priority macro F0.5 **0.7467 vs 0.2054**. It leads on all five corpora that can measure F2, with non-overlapping intervals.
- **It wins even on macro F2**, the recall-weighted metric picked because it favours the fusion's design. That was not the expected outcome.
- **The fusion's only win is recall, and it buys it by firing on everything** — prediction rate 0.9999, document specificity 0.0006. It passes the 0.90 per-tag recall gate degenerately.
- **Under the declared headline policy the winner is arm-C**, because the cascade fails the 0.90 recall gate (25/55 pairs) and the fusion does not. **Under the contra-view policy nothing ships.** Neither model is deployable as it stands.

- **The read window is not the story.** Arm C reads 12× more of every document than arm A and gains **+0.0009 macro F2**. The control ruled itself out.
- **Nothing was promoted, packaged or deployed** — this was a measurement run.

Date	2026-08-25
Author	Ryan Lence
Project	projects/pii-head-to-head-v1
Branch	main
Run ID	H1
Scope	two lineages re-tuned on the full 1-t train; 3 arms × 8 sealed corpora = 24 measured results
Request	<i>"do a head to head on both models. train them on this full dataset ... then eval them on each of the datasets in eval ... record all the metrics in the log F1, F3, micro_F1, F0.5, F2, precision, recall, etc.. also make sure to record the results of each pii tag as well from the eval each run needs to be recorded in the log"</i>
Outcome	Cascade wins on quality by a wide margin; no arm clears both declared policies; nothing promoted

Verification

The cached-feature path was proved identical to the serving path, not assumed. Every arm re-scored 96 sampled documents by reading the real file and calling the model's own `predict()`: **0 mismatches on all three arms.**

That check earned its place. On its first run it failed arm B with 2 of 96 disagreeing, all .docx. The cause is real and worth recording: `_xml_archive_text` concatenates zip-XML parts until it has `limit` characters, so for Office files the **extraction** — not just the truncation — depends on the limit. One datax file yields 10,788 characters at `limit=12000` and 11,995 at `limit=20000`, diverging in the tail. Measured scope:

Corpus	Documents affected	Extensions
datax eval	564 of 4,000 (14.1%)	all .docx
govdocs2 eval	0 of 6,578 (0.00%)	—

Each arm is now verified — and was trained — at its own lineage's read limit (20,000 for the fusion, 12,000 for the cascade). That is the only internally consistent choice: arm B's gate and heads were *fitted* on the 12,000-limit extraction, so scoring against 20,000-limit text would measure a model on input it never saw.

Residual confound, stated rather than buried: on those 564 datax documents arm C sees slightly more tail text than arm B. It is confined to one of eight corpora, affects only the tail, and datax carries positive-only tag gold — so it can touch recall and document metrics but never a precision-bearing number.

The evaluator was checked against closed-form values before it produced a number (`training/h2h_eval_selftest.py`): all four F-betas against their closed forms and correctly ordered; macro

excluding tags with no gold; micro pooling correctly; every CI bracketing its point estimate; and the three properties that would otherwise flatter a model — precision-bearing metrics None on positive-only gold rather than 0.0, no document scope on a prevalence-1.0 corpus, and a sub-min_support tag scoped with None plus its support.

Reproduction fidelity. Retrained from scratch, arm A lands within 0.004 of the shipped pii-priority-fusion-1k-v1 on both its headline numbers (macro F2 0.4800 vs 0.4835; priority macro F0.5 0.2060 vs 0.2057) and passes the same 55 measurable gates. Arm B benches at 3.916 ms / 314 docs/s against the shipped cascade's 4.11 ms / 310 docs/s, and its materialisation reproduced its own trial with drift **0.00000** against a 0.02 tolerance.

Artifacts

Path	What
reports/26-08-25_head-to-head.md	this report
reports/26-08-25_head-to-head.pdf	rendering of it
reports/26-08-25_head-to-head.commands.txt	complete unabridged command history
reports/26-08-25_Experiment-Log.xlsx	24 log rows, per-metric CIs, per-tag tab, data-quality tab
reports/_tables.md	every generated table, including the full per-tag worst-first listings
run.json	24 arms, 24 commands, both decisions, data quality
decision/headline.json, decision/precision_view.json	both recorded DecisionResults
evaluations/arm_{A,B,C}.json	model-level arms with all scopes and CIs
evaluations/latency_{A,B,C}.json	one-core bench evidence
models/{cascade,fusion_1000,fusion_12000}/	the three trained artifacts
tuning/*/trials.json	all 2,000 trials
mlflow.db	every trial as a tracked run under pii-head-to-head-v1

Command output is in the .commands.txt, not in this report.

Model scoreboard

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Macro F1	Docs/s ec	ms/d oc
fusion-1k	10360_betterdataai_ner_silver_eval_10.36k				0.208	1360.8	0.7
fusion-1k	10626_ai4privacy_pii_masking_eval_10.63k				0.424	1360.8	0.7
fusion-1k	20000_pii_holdout_20.00k				0.477	1360.8	0.7
fusion-1k	30000_pii2_eval_25.15k				0.413	1360.8	0.7

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
fusion-1k	38937_openpii_pii_eval_38.94k				0.510	1360.8	0.7
fusion-1k	4000_datax-dualjudge-evalset-1.32k					1360.8	0.7
fusion-1k	5617_nemotron_eval_5.36k					1360.8	0.7
fusion-1k	6589_govdocs2-dualjudge-eval20-3.53k					1360.8	0.7
steady-cascade	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
steady-cascade	10626_ai4privacy_pii_masking_eval_10.63k				0.729	314.1	3.2
steady-cascade	20000_pii_holdout_20.00k				0.709	314.1	3.2
steady-cascade	30000_pii2_eval_25.15k				0.757	314.1	3.2
steady-cascade	38937_openpii_pii_eval_38.94k				0.657	314.1	3.2
steady-cascade	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
steady-cascade	5617_nemotron_eval_5.36k					314.1	3.2
steady-cascade	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
fusion-12k	10360_betterdataai_ner_silver_eval_10.36k				0.208	270.3	3.7
fusion-12k	10626_ai4privacy_pii_masking_eval_10.63k				0.425	270.3	3.7
fusion-12k	20000_pii_holdout_20.00k				0.485	270.3	3.7
fusion-12k	30000_pii2_eval_25.15k				0.390	270.3	3.7
fusion-12k	38937_openpii_pii_eval_38.94k				0.511	270.3	3.7
fusion-12k	4000_datax-dualjudge-evalset-1.32k					270.3	3.7
fusion-12k	5617_nemotron_eval_5.36k					270.3	3.7
fusion-12k	6589_govdocs2-dualjudge-eval20-3.53k					270.3	3.7

Measured 2026-08-25 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.